Siemens FTGS Track Circuit Receiver Voltage Thresholds

Normal Voltage Range, Ballast Diagnostics, and Fault Investigation

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1. FTGS Track Circuit Overview

The Siemens FTGS (Frequency-shift Track Circuit with Guard Signal) operates by transmitting an audio-frequency signal through the rail and measuring the received voltage at the far end. The voltage received depends on the resistance of the ballast and rail path. A high ballast resistance reduces leakage and results in a higher received voltage. A low ballast resistance (due to contamination or moisture) increases leakage, reducing receiver voltage. When receiver voltage drops below the occupancy threshold, the system reports the section as occupied regardless of whether a train is present.

[Schematic: FTGS track circuit transmission and receiver configuration]

Figure: FTGS track circuit transmission and receiver configuration

2. Receiver Voltage Specifications

Condition	Receiver Voltage	Interpretation / Action
Normal - unoccupied	1.8 - 2.4 V	Track clear; no action
Normal - occupied (train shunt)	< 0.15 V	Train present; normal operation
Low - monitor	1.2 - 1.8 V	Monitor trend; inspect ballast within 14 days
Critical - false occupancy risk	< 1.2 V	Investigate immediately; inspect ballast
Transmitter or bonding fault	< 0.5 V (no train)	Check transmitter output and bonding wires

WARNING: A receiver voltage below 1.2 V with no train present will trigger a false occupancy signal. This may cause unnecessary train stops and delay. Notify the control centre and carry out a ballast inspection within 24 hours.

3. Identifying the Failure Mode from Voltage Trend

The shape of the voltage decline provides important diagnostic information:

- Gradual decline over weeks (more than 0.1 V per week consistently): Typically indicates progressive insulated joint degradation (cross-reference MAN-TC-001) or slowly increasing ballast resistance from contamination. Schedule IRJ resistance measurement and visual ballast inspection.
- Sudden drop over 2-5 days to below 1.2 V: Likely ballast contamination from a vegetation overgrowth event, flooding, or track maintenance that disturbed the ballast profile. Carry out urgent visual inspection and vegetation clearance.
- Intermittent dips followed by recovery: Possible loose bonding wire or intermittent IRJ contact. Inspect all bonding connections along the section.
- Stable low voltage with no trend: Transmitter output level may have drifted. Measure transmitter output and adjust if outside specification.

4. Ballast Inspection Procedure

A visual ballast inspection must be carried out by a track geometer or P-way technician. Key items to assess and clear:

- Vegetation (grass, weeds) growing through ballast within the section - cut and treat.
- Mud or fine material contamination of the ballast voids - ballast cleaning or replacement required.

- Water pooling or poor drainage adjacent to the track - raise a drainage defect order.
- Evidence of recent track maintenance (tamping, stoneblowing) that may have disturbed ballast continuity - allow 48 hours for ballast to settle and re-test voltage.

> *NOTE: After clearance, allow one wet-weather cycle (rainfall event) before concluding the ballast resistance has returned to normal. Continue daily voltage monitoring for at least 7 days post-clearance before closing the defect.*